

RITIKA SHRESTHA

Data Science and Intelligent Systems

Contact: +33 7 83 48 41 33 · ritika.shrestha@telecom-sudparis.eu
Socials: linkedin.com/in/ritikashrestha/ · github.com/RitikaShrestha-me ·
Availability: 6-month end-of-study internship from February 2027 · International student (VLS-TS)
Languages: English (C1) · French (Beginner, A1)

CAREER PROFILE

Master's student in Data Science and Network Intelligence at Télécom SudParis, graduating in March 2028, with a computer science background and hands-on experience in data analysis, machine learning, software development, and IoT systems. Experienced in building data pipelines, monitoring dashboards, and data-driven applications, with a strong interest in intelligent systems, performance optimization, and turning complex data into actionable insights. Comfortable collaborating with cross-functional teams in Agile environments. Seeking a 6-month end-of-study internship in data science, data analytics, or intelligent systems engineering.

EDUCATION

Télécom SudParis – Institut Polytechnique de Paris, France
2026 – present

M2 in Data Science and Network Intelligence

Glacial Lake Outburst Flood (GLOF) Early Warning System | IoT & Digital Twins

- Designed a prototype early-warning system for glacial lake floods: weighted hazard score combined with population and distance exposure into Low / Moderate / High risk.
- Built the data pipeline: synthetic sensor streams over MQTT into InfluxDB, validated and scored by a modular n8n workflow in JavaScript (range checks, missing-value handling, sensor-health flags, timestamp alignment).
- Automated alerting: critical alerts to authorities via Telegram and SMS, operator review for warnings, a sensor watchdog for sensor failures, and a live Next.js dashboard (risk state, sensor health, alert history).
- Specified 5 requirements with C4/UML diagrams; designed an edge fail-safe (local siren); tested 5 scenarios.

Technologies: MQTT, Eclipse Mosquitto, InfluxDB, n8n, JavaScript, Google Sheets, Docker, Next.js, Telegram, SMS8, C4/UML

Asian Institute of Technology, Thailand
2025 – present

M.Eng in IoT Systems Engineering (M1) | First year: Grade A (3.94/4) | HM King's Scholarship recipient

IoT Ranger | IoT Systems Project

- Built an IoT monitoring and device management platform inspired by NETPIE and AWS IoT: MQTT (EMQX broker) and HTTP device connectivity, real-time telemetry over WebSockets, and device shadow / live-feed synchronization.
- Designed a modular FastAPI backend (data pipeline, configuration, device registry, historical-data and WebSocket services) with JWT and Firebase authentication (email/password, Google) and broker-level device authentication.
- Stored the latest values and 5-minute history in MongoDB (device registry, user accounts, dashboard configs) and 1-second sensor readings in InfluxDB (7-day retention); built a SvelteKit app with a custom dashboard builder, alerts and Grafana monitoring.

Technologies: FastAPI, SvelteKit, MQTT, EMQX, WebSockets, MongoDB, InfluxDB, Grafana, Firebase Authentication, JWT, REST APIs, Docker

AI-Assisted Wireless User Scheduling | Machine Learning

- Trained a Random Forest on 20,000 Monte Carlo samples across 15 features to emulate Proportional Fair scheduling, achieving 92.95% test accuracy with grid search and 5-fold cross-validation.
- Benchmarked against PF, Round Robin and Max-SNR on 4,000 samples, achieving near-PF throughput (within 0.5%), 40% higher throughput than Round Robin, and 1.8× Max-SNR fairness; validated a capacity model showing 23–52% multiuser diversity gains.

Technologies: Python, Random Forest, Monte Carlo simulation, grid search, cross-validation

Visvesvaraya Technological University (VTU), India
2018 – 2022

B.E. in Computer Science & Engineering | 8.57/10 CGPA

WORK EXPERIENCE

Viveka Services – Associate Frontend Developer
Nov 2023 – Feb 2025

- Delivered eligibility, contract and employee-data features for a US health insurance platform serving 450K+ users, collaborating across backend, QA and product teams.
- Implemented Single-SPA micro-frontends and a shared WCAG 2.1 AA component library for Eligibility, reducing bundle size by ~25%.
- Built FindCare dashboards to compare healthcare provider prices and visualize cost data, helping users make informed decisions.
- Optimized FindCare using Next.js SSR, reducing page load times by 30%.

Technologies: React, TypeScript, Next.js, Single-SPA, Redux, React Query, SWR, Material UI, Styled-components, Webpack, Vite, GitLab CI, Jest, React Testing Library, Jira

BalooChat Technology – Software Development Engineer
Sept 2022 – Apr 2023

- Built the mobile chat app UI in React Native, following best practices to deliver an intuitive user experience.
- Worked with the design team to build a scalable, maintainable codebase from reusable components and efficient caching.
- Integrated REST API data with backend developers, improving app performance and load times.

Technologies: React Native, JavaScript/TypeScript, REST APIs

TaxAdda Private Limited – React Frontend Developer
June 2021 – Dec 2021

- Built a seller dashboard from scratch with user authentication, integrating backend APIs for each action users perform.
- Built scalable workflows to import, filter, process, and transfer transaction data between tables across multiple online selling platforms.
- Implemented JSON download for tables and added Google reCAPTCHA to block bot access to the website API.

Technologies: React, JavaScript/TypeScript, REST APIs, Google reCAPTCHA

Additional projects and work experience are described in detail on my LinkedIn profile: linkedin.com/in/ritikashrestha/